

Veer Suraksha Grid

Institutional Research Brief & Competitive Analysis

A dual-mode IoT + web platform designed to protect sanitation workers of Solapur Municipal Corporation from hazardous working conditions.

LIVE PROTOTYPE

veer-suraksha-grid.netlify.app

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Technology-Enabled Sanitation Worker Safety

HACKATHON

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01 SECTION 1 — SMC INSTITUTIONAL FIT

Core Argument: Veer Suraksha Grid does not ask SMC to replace anything. It plugs into, extends, and makes smarter what SMC already operates.

1A. SMC GIS Portal — smcgis.solapurcorporation.org

What exists:

SMC operates a live, publicly accessible GIS dashboard at smcgis.solapurcorporation.org. It is built on CyberTech's **GeoCivic® platform** — a cloud-based, modular system designed for Urban Local Bodies to add layers incrementally. Confirmed live data layers include:

- Sewer_Pipeline_Network
- Chamber_Manhole ← most critical for VeerRegistry
- Sewerage_Collection_Point
- Water network (Pipeline_Network, Storage_Tank, WTP, SMC_Bore)
- Road network & Administrative boundary layers

How VeerRegistry plugs in:

The GIS portal already has a **Chamber_Manhole layer**. VeerRegistry enriches it — every manhole entry gets: GPS coordinates, last inspection date, gas readings history, VeerRisk classification (Red/Yellow/Green), and HC-SR04 depth measurements. This is a **data feed into an existing layer**, not a new portal. SMC's GIS department gets richer asset data; we get geographic display.

"SMC already has a manhole layer on its GIS portal. We make it smarter — every manhole gets a risk score, an inspection history, and a pre-entry gas reading. Same portal, better data."

1B. mysolapur App & SMC E-Services Portal

What exists:

- **mysolapur app** (Android, com.smc.mysolapur) — officially promoted on SMC homepage and @smcsolapur Instagram (9,960+ followers). Covers e-services of SMC.
- **E-Services Portal** (eservices.solapurcorporation.org) — property tax, birth/death certificates, complaint lodging, and application status tracking.
- **Note for Q&A**: App reviews cite 'complaint section does not work' — an argument in our favour. The grievance pipeline is broken and we fix it.

How Grievance Integration plugs in:

Complaints on mysolapur or the e-services portal related to blocked drains or manholes feed into our module, which automatically generates a **VeerProbe inspection task**, a **VeerRisk trigger**, and a **task-completion webhook** that closes the original complaint once work is done. No new channel is created — the existing channel gets an automated responder.

1C. Ghantagadi GPS Fleet — The Existing Tracking Parallel

SMC's homepage prominently links a **Ghantagadi Information** section confirming the garbage collection fleet is GPS-tracked and citizen-visible. SMC has already built institutional muscle for GPS-based fleet tracking.

How FleetEye plugs in:

FleetEye applies the same concept to the **sanitation response fleet** (jetting machines, sludge tankers) instead of garbage trucks. When a VeerPass request is received, FleetEye checks that a mechanized vehicle has been dispatched and logged its arrival. Only then does VeerPass unlock manual-entry authorization.

"SMC already tracks Ghantagadi vehicles by GPS. FleetEye is the same idea for the sewer cleaning fleet — and it creates proof that mechanization was attempted before a worker went in."

1D. SMC Sanitation Fleet & NAMASTE Scheme

Under the **NAMASTE scheme** (National Action for Mechanised Sanitation Ecosystem, July 2023 — ₹349.73 crore, MoSJE + MoHUA), all ULBs including SMC are mandated to:

- Profile all Sewer and Septic Tank Workers (SSWs) with a unique NAMASTE ID
- Procure mechanized cleaning equipment (jetting, suction, robotic) via capital subsidy under Swachhhata Udyami Yojana (SUY)
- Set up Sanitation Response Units (SRUs) with safety devices

Maharashtra's **Manhole to Machinehole scheme** (August 2022, CM Eknath Shinde) targets mechanization across 11 major municipal corporations, providing state-level policy backing for FleetEye's machine-first enforcement logic.

Typical fleet for SMC's city scale:

- High-pressure jetting/rodding machines (pipeline blockage clearance)
- Sludge/vacuum suction tankers (septic tank and wet sewer cleaning)
- Septic tank emptying vehicles
- Potentially a robotic sewer inspection unit (NAMASTE/Smart Cities funding)

1E. SCDCL, ICCC & the SwachhCommander Extension

Solapur was selected in **Round 1 of the Smart Cities Challenge** and operates through **Solapur City Development Corporation Limited (SCDCL)**. Per Ministry of Housing and Urban Affairs (PIB 2023), ICCCs are operational in all 100 smart cities including Solapur, covering solid waste, traffic, water, and safety. SCDCL completed projects include sanitation improvements; ongoing projects include sewerage system upgrades.

What the ICCC currently does NOT have:

- Real-time underground sewer gas telemetry (CH₄, CO, LPG)
- Worker biometric monitoring (heart rate, GPS position, SOS status)
- Pre-entry inspection dashboards (gas readings before human descent)

- Digital authorization chains with immutable audit trails

How SwachhCommander fits:

SwachhCommander is a **dedicated sanitation safety vertical** that displays as a single screen within the ICCC's existing dashboard via a REST API or webhook from our Firebase/InfluxDB backend. ICCC operators do not change workflows. In a sewer emergency, the SwachhCommander alert appears in the ICCC's incident feed — no new building, no new control room.

"The ICCC monitors the city. SwachhCommander monitors what happens 20 feet underground when a worker descends. Complementary, not competing. We ask for one screen — not a new building."

1F. SMC's 8 Administrative Zones — Making VeerRisk Credible

SMC administers Solapur across **8 administrative zones** and 102 wards, covering 178.57 sq. km with ~12.27 lakh residents (2021). Population is relatively uniformly distributed (avg ~36,599 per ward, SD ~3,184).

Sewer risk relevance by zone type:

Zone Type	Primary Risk Factor	Key Sensor
Old city core (textile mill belt)	Pre-independence sewers; methane from organic settlement	MQ-4 (CH4)
Industrial corridor zones	Industrial effluent; chemical gas beyond standard sewer gases	MQ-2 (LPG/smoke)
Dense slum / peripheral zones	Low mechanization; high worker-to-machine ratio	All sensors + SOS

1G. SMC Grievance Channels — End-to-End Workflow

Channel	Status	Integration Point
mysolapur app	Active / Official	Sewer complaints trigger VeerProbe task auto-creation
eservices.solapurcorporation.org	Active	Complaint form feeds task pipeline; completion closes ticket
Aaple Sarkar / PGR Maharashtra	State-level	Escalation receiver; module pushes task-completion status back
SMC Social Media (@smcsolapur)	Active (HC-mandated)	Monitor manual scavenging reports per Bombay HC Aug 2024 order

Complete end-to-end grievance loop:

1. Citizen reports blocked sewer on mysolapur app → complaint ID generated
2. Grievance Integration detects 'sewer' keyword, pulls complaint coordinates
3. Auto-creates a VeerProbe inspection task, assigns to nearest field team
4. VeerProbe pre-entry gas check completed → reading logged in VeerBlackBox
5. If safe → VeerPass authorization issued → worker dispatched

6. On completion → Grievance Integration pushes status update → complaint resolved in portal
7. Full chain is timestamped and immutable in VeerBlackBox

02 SECTION 2 — LEGAL COMPLIANCE AS COMPETITIVE WEAPON

Core Argument: SMC is under active legal pressure from the Supreme Court and Bombay High Court. Veer Suraksha Grid gives them the evidence trail they need to demonstrate compliance — and the legal defense they need if something goes wrong.

2A. PEMSAR Act 2013 — Specific Sections Creating Municipal Liability

Rule 3(1)(e)	→ VeerPass	Manual entry only permitted after the CEO of the local authority has permitted it in writing, recording specific valid reasons. VeerPass digitizes this process — the authorized officer issues a timestamped digital permit logged permanently in VeerBlackBox.
Rule 4	→ Hardware (MQ-2+MQ-4+MQ-7+DHT11)	Employers must provide a 4-gas monitor to every worker entering a sewer. Our sensor combination covers all four required parameters: LPG/smoke (MQ-2), methane (MQ-4), CO (MQ-7), and temperature/humidity (DHT11).
Rule 6(3) Safety Precautions	→ VeerProbe	Before entry, a supervisor must inspect whether the sewer serves nearby industries to anticipate chemical hazards. VeerProbe — lowered into the manhole before human entry — is the technological fulfillment of this rule.
Rule 3(2)	→ FleetEye	Before entry, sewage must be totally emptied where possible. FleetEye's machine-first enforcement creates a verifiable record that mechanized emptying was attempted first — the proof Rule 3(2) requires.

2B. Dr. Balram Singh v. Union of India (Supreme Court, Oct 20, 2023)

- A bench of Justices S. Ravindra Bhat and Aravind Kumar issued **14 binding directions**. Key mandates:
- Complete eradication of manual sewer cleaning must be pursued in a phased manner by all municipalities
 - Compensation for sewer deaths increased to **■30 lakhs per family** (up from **■10 lakhs** under 2014 order)
 - Governments must devise accountability mechanisms especially where sewer deaths have occurred
 - Outsourced work must not require sewer entry under any circumstances
 - Court explicitly noted no municipality had achieved full compliance even after the NAMASTE scheme launch

How VeerPass + VeerBlackBox respond:

VeerBlackBox creates an immutable incident log — who authorized entry, what gas readings were at time of entry, which machine was deployed first. This is the **accountability mechanism** the Court demanded. If SMC faces a sewer death inquiry, a judge will ask three questions: 'Was authorization given? Was gas checked? Was a machine sent first?' VeerBlackBox answers all three with timestamps.

2C. Bombay High Court 2024 — Active Oversight on Maharashtra Municipalities

In **Shramik Janata Sangh v. State of Maharashtra** (SCC OnLine Bom 1202 & 2745; orders April and August 2024; Justices Nitin Jamdar and M.M. Sathaye):

- HC directed compliance information from major Maharashtra corporations in phase 1 — Solapur is in the expansion scope for phase 2 scrutiny
- Court noted Maharashtra had not framed its own PEMSR rules — Central PEMSR Rules 2013 apply directly, including Rule 3(1)(e) written authorization mandate
- Court ordered the Department of Social Welfare to establish dedicated email and social media handles per district vigilance committee to receive manual scavenging reports
- State's nodal officer admitted instances of manual scavenging in April and August 2024 despite claiming Maharashtra was scavenging-free — making state self-certification legally untenable

The risk to SMC: The Bombay HC is actively expanding scrutiny to more Maharashtra corporations. Without a traceable authorization system, SMC has no compliance defense when the court turns its attention to Solapur.

2D. Financial Exposure — The ROI Argument for the Commissioner

Event	Compensation	Source
Sewer worker death	■30 lakhs per family	SC, Balram Singh Oct 2023
Pre-2023 deaths (since 1993)	■10 lakhs per death	SC Order March 2014; 948 families compensated nationally
Criminal liability on local authority CEO	Imprisonment + fine	PEMSR Act 2013, Sections 8-9

Between 2018-2022, **339 workers died** cleaning sewers nationally. Maharashtra is among the highest-incident states. Even one death in Solapur = ■30L mandatory payout + legal costs + reputational damage + personal criminal liability for the Municipal Commissioner. The platform cost of Veer Suraksha Grid is a fraction of one compensation payout.

2E. FleetEye — Machine-First Enforcement as Legal Defense

PEMSR Rules Rule 3(1): Manual entry is only permitted when mechanical equipment *cannot* be operated. FleetEye creates a **logged, timestamped record** that a jetting machine or suction vehicle was dispatched and arrived before VeerPass was issued.

Without FleetEye: SMC can claim it tried machines — but cannot prove it. With FleetEye: The dispatch log is the proof. This is the difference between a legal defense that holds and one that collapses under cross-examination.

03 SECTION 3 — COMPETITOR GAP ANALYSIS

Dimension	Safe Sphere	CODE XA	Logic Lords	IoTians	Veer Suraksha Grid	Consequence if Missing
(a) Pre-entry probe mode before descent	✗	✗	✗	✗	✓	Most sewer deaths occur in first 60 sec. No probe = worker enters blind.
(b) Digital authorization + audit trail	✗	✗	✗	✗	✓	PEMSR Rule 3(1)(e) violation. No legal defense if worker dies.
(c) Machine-first enforcement layer	✗	✗	✗	✗	✓	No proof mechanization was attempted. Liability shifts to municipality.
(d) Municipal command-center dashboard	✗	~	✗	~	✓	No city-level awareness. Devices isolated without coordination.
(e) AI-assisted hazard decision support	✗	✗	✗	✗	✓	Decision left to under-trained supervisor. Human error under stress.
(f) Multi-gas: CO + CH ₄ + LPG (all 3)	~1	~1	~1	?	✓	CO is colorless/odorless — kills silently. Single-sensor teams miss 2 of 3 lethal gases.
(g) Complaint-to-task workflow	✗	✗	✗	✗	✓	Citizen reports don't trigger inspection. Reactive-only response.
(h) Marathi-language interface	✗	✗	✗	✗	✓	Low-literacy workers cannot use English dashboards. Safety alerts ignored.
(i) Incident audit trail	✗	✗	✗	✗	✓	No accountability after a death. Compensation claims disputed without records.
(j) Risk zone classification (R/Y/G)	✗	✗	✗	✗	✓	All locations treated equally. High-risk manholes get same response as low-risk ones.

~ = partial/basic · ? = insufficient info · 1 = single MQ sensor only

Three Non-Negotiable Differentiators

01

Dual-Mode Hardware: Probe + Wearable

Every competing team built a wearable. Only we ask: 'What if you could check before anyone goes in?' VeerProbe mode — lowering the ESP32 device into the manhole before human entry — addresses the most critical gap. Most sewer deaths happen within the first 60 seconds of descent. Probe mode eliminates that 60-second window entirely.

02

Legal Compliance Infrastructure: VeerPass + VeerBlackBox

No competitor has thought about what happens after an incident. We have. VeerBlackBox gives SMC a timestamped, immutable authorization record. In a Bombay HC inquiry or Supreme Court compliance review, this record is the difference between compliance and contempt.

03

Municipal Integration Depth

Our system is designed to work within SMC's existing GIS portal (Chamber_Manhole layer), ICCC, Ghantagadi tracking ecosystem, and mysolapur grievance pipeline — not alongside them. No other team has mapped their solution to SMC's specific digital infrastructure. That is the difference between a proof-of-concept and a deployable system.

Sources

- PEMS Rules 2013 — ielrc.org/content/e1314.pdf
- Dr. Balram Singh v. Union of India — indiankanoon.org/doc/4950148/ | clpr.org.in
- Shramik Janata Sangh v. State of Maharashtra — SCC OnLine Bom 1202 (Apr 2024) & 2745 (Aug 2024)
- SMC GIS Portal — smcgis.solapurcorporation.org (Sewerage, Water, Admin layers)
- SMC Official Website — solapurcorporation.gov.in | Instagram @smcsolapur
- NAMASTE Scheme — socialjustice.gov.in/schemes/37 | PIB prid=2012373
- Solapur Municipal Corporation — Wikipedia | Grokipedia
- SCDCL Smart Cities — smartcities.gov.in/node/122 | PIB prid=1947455
- Maharashtra Manhole to Machinehole — The Week, August 2025